

GENERAL INSTRUCTIONS:

1. This paper contains 15 Multiple Choice Questions divided into 3 Sections: Physics, Chemistry & Mathematics.
2. Each question has four options (A, B, C, D) with exactly one correct option. Darken the corresponding circle.

Section A — Physics

1. What is the SI unit of electrical capacitance in terms of base units?

- (A) Farad (C / V) (B) Henry (Wb / A)
(C) Siemens (A / V) (D) Weber (J / A)

2. A particle moves along a circular path of radius R with constant speed v. What is the magnitude of average acceleration over half a revolution?

- (A) $2v^2 / (\pi * R)$ (B) v^2 / R
(C) $\pi * v^2 / (2 * R)$ (D) Zero

3. Light of wavelength 600 nm is incident normally on a diffraction grating with 5000 lines per cm. What is the maximum number of diffraction orders observed?

- (A) 2 (B) 3 (C) 5 (D) 7

4. Consider an idealized thermodynamic Carnot engine operating between temperatures $T_1 = 800$ K and $T_2 = 400$ K. In each cycle, the working substance absorbs 1000 Joules of thermal energy from the high-temperature reservoir. Calculate the theoretical thermal efficiency and the net mechanical work output delivered per cycle:

- (A) Efficiency = 50%, Work Output = 500 Joules
(B) Efficiency = 40%, Work Output = 400 Joules

- (C) Efficiency = 60%, Work Output = 600 Joules
(D) Efficiency = 75%, Work Output = 750 Joules

5. Which fundamental law of physics establishes that magnetic monopoles do not exist in isolation?

- (A) Gauss's Law for Magnetism ($\text{div } \mathbf{B} = 0$)
(B) Ampere's Circuital Law
(C) Faraday's Law of Electromagnetic Induction
(D) Coulomb's Law of Electrostatics

Section B — Chemistry

6. What is the IUPAC systematic name for the compound $\text{CH}_3\text{-CH}(\text{CH}_3)\text{-CH}_2\text{-CHO}$?

- (A) 2-Methylbutanal
(B) 3-Methylbutanal
(C) 3-Methylbutanoic acid
(D) 2-Methylbutan-1-one

7. Which of the following elements possesses the highest first ionization enthalpy?

- (A) Helium (He) (B) Fluorine (F) (C) Neon (Ne)
(D) Nitrogen (N) (E) Oxygen (O)

8. What is the oxidation state of Chromium (Cr) in the dichromate ion $[\text{Cr}_2\text{O}_7]^{2-}$?

- (A) +3 (B) +4 (C) +6 (D) +7

9. For a first-order chemical reaction with rate constant $k = 0.0693 \text{ min}^{-1}$, what is the half-life?

- (A) 5.0 minutes (B) 10.0 minutes
(C) 15.0 minutes (D) 20.0 minutes

10. Which coordination geometry is typically associated with d8 transition metal complexes with strong field ligands?

- (A) Tetrahedral (B) Square Planar
(C) Octahedral (D) Trigonal Bipyramidal

Section C — Mathematics

11. If α and β are the roots of quadratic equation $x^2 - 7x + 12 = 0$, find the value of $(\alpha^2 + \beta^2)$:

- (A) 25 (B) 37 (C) 49 (D) 13

12. Evaluate the definite integral: Integral from 0 to $\pi/2$ of $[\sin(x) / (\sin(x) + \cos(x))]$ dx

- (A) $\pi / 4$ (B) $\pi / 2$ (C) 1 (D) 0

13. What is the value of limit as x approaches 0 for: $(e^{3x} - 1) / (\sin(2x))$?

- (A) 1.0 (B) $1.5 (3/2)$ (C) 2.0 (D) $0.67 (2/3)$

14. If a matrix A of order 3×3 has determinant $\det(A) = 4$, what is the determinant of $2 * \text{adj}(A)$?

- (A) 64 (B) 128 (C) 256 (D) 512

15. Two fair 6-sided dice are thrown simultaneously. What is the probability that the sum is at least 10?

- (A) $1 / 12$ (B) $1 / 6$ (C) $1 / 4$ (D) $5 / 36$

Official Solutions & Answer Key

Master evaluation grid for automated reconciliation.

QUESTION NUMBER	SUBJECT / SECTION	CORRECT OPTION
Question 1	Section A - Physics	(A)
Question 2	Section A - Physics	(A)
Question 3	Section A - Physics	(B)
Question 4	Section A - Physics	(A)
Question 5	Section A - Physics	(A)
Question 6	Section B - Chemistry	(B)
Question 7	Section B - Chemistry	(A)
Question 8	Section B - Chemistry	(C)
Question 9	Section B - Chemistry	(B)
Question 10	Section B - Chemistry	(B)
Question 11	Section C - Mathematics	(A)
Question 12	Section C - Mathematics	(A)
Question 13	Section C - Mathematics	(B)
Question 14	Section C - Mathematics	(B)
Question 15	Section C - Mathematics	(B)